

TUGAS PRAKTIKUM STRUKTUR DATA



Oleh:

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2511533005

MATA KULIAH STRUKTUR DATA

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Membuat program Algoritma Sorting dengan tema “Pengurutan Nama Mahasiswa Secara Alfabetis Menggunakan GUI”. Program dikembangkan menggunakan Java GUI (Swing) untuk menampilkan proses pengurutan data mahasiswa secara interaktif.

- Kode program java
 - a. Class Mahasiswa

```
1 package StrukturData;
2
3 public class Mahasiswa_2511533005 {
4     private String nama_3005;
5     private String nim_3005;
6     private String prodi_3005;
7
8     public Mahasiswa_2511533005(String nama_3005, String nim_3005, String prodi_3005) {
9         this.nama_3005 = nama_3005;
10        this.nim_3005 = nim_3005;
11        this.prodi_3005 = prodi_3005;
12    }
13
14    public String getNama_3005() {
15        return nama_3005;
16    }
17    public void setNama_3005(String nama_3005) {
18        this.nama_3005 = nama_3005;
19    }
20    public String getNim_3005() {
21        return nim_3005;
22    }
23    public void setNim_3005(String nim_3005) {
24        this.nim_3005 = nim_3005;
25    }
26    public String getProdi_3005() {
27        return prodi_3005;
28    }
29    public void setProdi_3005(String prodi_3005) {
30        this.prodi_3005 = prodi_3005;
31    }
32
33    @Override
34    public String toString() {
35        return nama_3005 + " - " + nim_3005 + " - " + prodi_3005;
36    }
37
38
39
40
```

- b. Class Musik

```

1 package StrukturData;
2
3 import java.awt.BorderLayout;
4 import java.awt.FlowLayout;
5 import java.awt.Font;
6 import java.util.ArrayList;
7
8 import javax.swing.BorderFactory;
9 import javax.swing.JButton;
10 import javax.swing.JComboBox;
11 import javax.swing.JFrame;
12 import javax.swing.JLabel;
13 import javax.swing.JOptionPane;
14 import javax.swing.JPanel;
15 import javax.swing.JScrollPane;
16 import javax.swing.JTable;
17 import javax.swing.JTextArea;
18 import javax.swing.JTextField;
19 import javax.swing.table.DefaultTableModel;
20
21 public class SortingGUI_2511533005 extends JFrame {
22
23     private static final long serialVersionUID = 1L;
24
25     // ArrayList mahasiswa
26     private ArrayList<Mahasiswa_2511533005> listMahasiswa_3005 = new ArrayList<>();
27
28     // Komponen GUI
29     private JTextField txtNama_3005;
30     private JTextField txtNim_3005;
31     private JTextField txtProdi_3005;
32     private JButton btnTambah_3005;
33     private JButton btnHapus_3005;
34     private JButton btnSorting_3005;
35
36     private JComboBox<String> comboSorting_3005;
37
38     private JTable tableMahasiswa_3005;
39     private DefaultTableModel modelTable_3005;
40
41     private JTextArea txtArea_3005;
42
43     // Constructor
44     public SortingGUI_2511533005() {

```

```

43     // Constructor
44     public SortingGUI_2511533005() {
45
46         setTitle("Program Sorting Mahasiswa");
47         setSize(900, 500);
48         setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
49         setLocationRelativeTo(null);
50         setLayout(new BorderLayout());
51
52         JPanel panelInput_3005 = new JPanel(new FlowLayout());
53         panelInput_3005.add(new JLabel("Nama"));
54         txtNama_3005 = new JTextField(10);
55         panelInput_3005.add(txtNama_3005);
56
57         panelInput_3005.add(new JLabel("NIM"));
58         txtNim_3005 = new JTextField(10);
59         panelInput_3005.add(txtNim_3005);
60
61         panelInput_3005.add(new JLabel("Prodi"));
62         txtProdi_3005 = new JTextField(10);
63         panelInput_3005.add(txtProdi_3005);
64
65         btnTambah_3005 = new JButton("Tambah Data");
66         panelInput_3005.add(btnTambah_3005);
67
68         btnHapus_3005 = new JButton("Hapus Data");
69         panelInput_3005.add(btnHapus_3005);
70
71         add(panelInput_3005, BorderLayout.NORTH);
72
73         String[] kolom_3005 = {"Nama", "NIM", "Prodi"};
74         modelTable_3005 = new DefaultTableModel(kolom_3005, 0);
75         tableMahasiswa_3005 = new JTable(modelTable_3005);
76         JScrollPane scrollTable_3005 = new JScrollPane(tableMahasiswa_3005);
77
78         add(scrollTable_3005, BorderLayout.CENTER);
79
80         JPanel panelBawah_3005 = new JPanel(new FlowLayout());
81
82         comboSorting_3005 = new JComboBox<>();
83         comboSorting_3005.addItem("Insertion Sort");
84         comboSorting_3005.addItem("Selection Sort");
85         comboSorting_3005.addItem("Bubble Sort");

```

```

80      JPanel panelBawah_3005 = new JPanel(new FlowLayout());
81
82      comboSorting_3005 = new JComboBox<>();
83      comboSorting_3005.addItem("Insertion Sort");
84      comboSorting_3005.addItem("Selection Sort");
85      comboSorting_3005.addItem("Bubble Sort");
86
87      panelBawah_3005.add(new JLabel("Pilih Sorting"));
88      panelBawah_3005.add(comboSorting_3005);
89
90      btnSorting_3005 = new JButton("Mulai Sorting");
91      panelBawah_3005.add(btnSorting_3005);
92
93      add(panelBawah_3005, BorderLayout.SOUTH);
94
95      txtArea_3005 = new JTextArea(20, 40);
96      txtArea_3005.setEditable(false);
97      txtArea_3005.setFont(new Font("Monospaced", Font.PLAIN, 14));
98      txtArea_3005.setBorder(BorderFactory.createTitledBorder("Proses Sorting"));
99
100     JScrollPane scrollText_3005 = new JScrollPane(txtArea_3005);
101
102     add(scrollText_3005, BorderLayout.EAST);
103
104     btnTambah_3005.addActionListener(e_3005 -> tambahData_2511533005());
105
106     btnHapus_3005.addActionListener(e_3005 -> hapusData_2511533005());
107
108     btnSorting_3005.addActionListener(e_3005 -> prosesSorting_2511533005());
109 }
110 private void tambahData_2511533005() {
111     String nama_3005 = txtNama_3005.getText();
112     String nim_3005 = txtNim_3005.getText();
113     String prodi_3005 = txtProdi_3005.getText();
114
115     if (nama_3005.isEmpty() || nim_3005.isEmpty() || prodi_3005.isEmpty()) {
116         JOptionPane.showMessageDialog(this, "Data tidak boleh kosong!");
117         return;
118     }
119     Mahasiswa_2511533005 mhs_3005 =
120         new Mahasiswa_2511533005(nama_3005, nim_3005, prodi_3005);
121     listMahasiswa_3005.add(mhs_3005);
122
123

```

```

120     Mahasiswa_2511533005 mhs_3005 =
121         new Mahasiswa_2511533005(nama_3005, nim_3005, prodi_3005);
122     listMahasiswa_3005.add(mhs_3005);
123
124     modelTable_3005.addRow(new Object[] {
125         nama_3005,
126         nim_3005,
127         prodi_3005
128     });
129
130     txtNama_3005.setText("");
131     txtNim_3005.setText("");
132     txtProdi_3005.setText("");
133 }
134 private void hapusData_2511533005() {
135     int index_3005 = tableMahasiswa_3005.getSelectedRow();
136     if (index_3005 != -1) {
137         listMahasiswa_3005.remove(index_3005);
138         modelTable_3005.removeRow(index_3005);
139     } else {
140         JOptionPane.showMessageDialog(this, "Pilih data yang ingin dihapus!");
141     }
142 }
143 private void prosesSorting_2511533005() {
144     String pilihan_3005 =
145         comboSorting_3005.getSelectedItem().toString();
146     txtArea_3005.setText("");
147     if (pilihan_3005.equals("Insertion Sort")) {
148         insertionSort_2511533005();
149     } else if (pilihan_3005.equals("Selection Sort")) {
150         selectionSort_2511533005();
151     } else {
152         bubbleSort_2511533005();
153     }
154     updateTable_2511533005();
155 }
156 private void insertionSort_2511533005() {
157     txtArea_3005.append("=== INSERTION SORT ===\n");
158     for (int i_3005 = 1; i_3005 < listMahasiswa_3005.size(); i_3005++) {
159         Mahasiswa_2511533005 key_3005 =
160             listMahasiswa_3005.get(i_3005);
161         int j_3005 = i_3005 - 1;
162         while (j_3005 >= 0 &&
163             listMahasiswa_3005.get(j_3005).

```

```

157 private void insertionSort_2511533005() {
158     txtArea_3005.append("=== INSERTION SORT ===\n");
159     for (int i_3005 = 1; i_3005 < listMahasiswa_3005.size(); i_3005++) {
160         Mahasiswa_2511533005 key_3005 =
161             listMahasiswa_3005.get(i_3005);
162         int j_3005 = i_3005 - 1;
163         while (j_3005 >= 0 &&
164             listMahasiswa_3005.get(j_3005)
165                 .getNama_3005()
166                 .compareToIgnoreCase(key_3005.getNama_3005()) > 0) {
167             listMahasiswa_3005.set(
168                 j_3005 + 1,
169                 listMahasiswa_3005.get(j_3005));
170             j_3005--;
171         }
172         listMahasiswa_3005.set(j_3005 + 1, key_3005);
173         txtArea_3005.append(
174             "Langkah " + i_3005 + " : "
175             + listToString_2511533005() + "\n");
176     }
177 }
178 private void selectionSort_2511533005() {
179     txtArea_3005.append("=== SELECTION SORT ===\n");
180     for (int i_3005 = 0;
181         i_3005 < listMahasiswa_3005.size() - 1;
182         i_3005++) {
183         int minIndex_3005 = i_3005;
184         for (int j_3005 = i_3005 + 1;
185             j_3005 < listMahasiswa_3005.size();
186             j_3005++) {
187             if (listMahasiswa_3005.get(j_3005)
188                 .getNama_3005()
189                 .compareToIgnoreCase(
190                     listMahasiswa_3005.get(minIndex_3005)
191                     .getNama_3005()) < 0) {
192                 minIndex_3005 = j_3005;
193             }
194         }
195         Mahasiswa_2511533005 temp_3005 =
196             listMahasiswa_3005.get(i_3005);
197         listMahasiswa_3005.set(
198             i_3005,
199             listMahasiswa_3005.get(minIndex_3005));
200         listMahasiswa_3005.set(minIndex_3005, temp_3005);

```

```

201     }
202     Mahasiswa_2511533005 temp_3005 =
203         listMahasiswa_3005.get(i_3005);
204     listMahasiswa_3005.set(
205         i_3005,
206         listMahasiswa_3005.get(minIndex_3005));
207     listMahasiswa_3005.set(minIndex_3005, temp_3005);
208     txtArea_3005.append(
209         "Pass " + (i_3005 + 1) + " : "
210         + listToString_2511533005() + "\n");
211 }
212 private void bubbleSort_2511533005() {
213     txtArea_3005.append("=== BUBBLE SORT ===\n");
214     for (int i_3005 = 0;
215         i_3005 < listMahasiswa_3005.size() - 1;
216         i_3005++) {
217         for (int j_3005 = 0;
218             j_3005 < listMahasiswa_3005.size() - i_3005 - 1;
219             j_3005++) {
220             if (listMahasiswa_3005.get(j_3005)
221                 .getNama_3005()
222                 .compareToIgnoreCase(
223                     listMahasiswa_3005.get(j_3005 + 1)
224                     .getNama_3005()) > 0) {
225                 Mahasiswa_2511533005 temp_3005 =
226                     listMahasiswa_3005.get(j_3005);
227                 listMahasiswa_3005.set(
228                     j_3005,
229                     listMahasiswa_3005.get(j_3005 + 1));
230                 listMahasiswa_3005.set(
231                     j_3005 + 1,
232                     temp_3005);
233             }
234         }
235         txtArea_3005.append(
236             "Pass " + (i_3005 + 1) + " : "
237             + listToString_2511533005() + "\n");

```

```

230         j_3005 + 1,
231         temp_3005);
232     }
233 }
234 txtArea_3005.append(
235     "Pass " + (i_3005 + 1) + " : "
236     + listToString_2511533005() + "\n");
237 }
238 }
239 private void updateTable_2511533005() {
240     modelTable_3005.setRowCount(0);
241
242     for (Mahasiswa_2511533005 mhs_3005 : listMahasiswa_3005) {
243         modelTable_3005.addRow(new Object[] {
244             mhs_3005.getNama_3005(),
245             mhs_3005.getNim_3005(),
246             mhs_3005.getProdi_3005()
247         });
248     }
249 }
250 private String listToString_2511533005() {
251     String hasil_3005 = "[";
252     for (int i_3005 = 0;
253         i_3005 < listMahasiswa_3005.size();
254         i_3005++) {
255         hasil_3005 +=
256             listMahasiswa_3005.get(i_3005)
257             .getNama_3005();
258         if (i_3005 != listMahasiswa_3005.size() - 1) {
259             hasil_3005 += ", ";
260         }
261     }
262     hasil_3005 += "]";
263     return hasil_3005;
264 }
265 public static void main(String[] args) {
266     new SortingGUI_2511533005().setVisible(true);
267 }
268 }
269 }
270 }
271 }
272 }
273 }

```

- Contoh output

Program Sorting Mahasiswa

Nama NIM Prodi

Nama	NIM	Prodi
Hamdi	2511531009	Informatika
Octa	2511533005	Informatika
Saka	25115330005	Informatika

Proses Sorting

```

=== INSERTION SORT ===
Langkah 1 : [Octa, Saka, Hamdi]
Langkah 2 : [Hamdi, Octa, Saka]

```

Pilih Sorting

Program Sorting Mahasiswa

Nama NIM Prodi

Nama	NIM	Prodi
Hamdi	2511531009	Informatika
Octa	2511533005	Informatika
Saka	25115330005	Informatika

Pilih Sorting Selection Sort

Proses Sorting

```
=== SELECTION SORT ===  
Pass 1 : [Hamdi , Saka, Octa]  
Pass 2 : [Hamdi , Octa, Saka]
```

Program Sorting Mahasiswa

Nama NIM Prodi

Nama	NIM	Prodi
Hamdi	2511531009	Informatika
Octa	2511533005	Informatika
Saka	25115330005	Informatika

Pilih Sorting Bubble Sort

Proses Sorting

```
=== BUBBLE SORT ===  
Pass 1 : [Octa, Hamdi, Saka]  
Pass 2 : [Hamdi, Octa, Saka]
```